Xianglong Tang

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SUMMARY

Senior Ground Software Engineer with 7+ years of experience architecting and delivering production-critical, web-based ground software platforms for spacecraft and satellite constellation operations. Deeply proficient in Python, FastAPI, TypeScript/Angular, and PostgreSQL, with a proven track record of deploying robust microservice architectures and distributed telemetry pipelines. Hands-on experience implementing CCSDS space packet protocols and Kubernetes-orchestrated ground data systems.

TECHNICAL SKILLS

Languages: Python, Go, Rust, TypeScript, C++, SQL | **Frameworks/Web:** Angular, FastAPI, Playwright E2E, REST, gRPC
Data/Infrastructure: PostgreSQL, InfluxDB, Kubernetes, Docker, Kafka, Redis, CI/CD, Git
Space Standards: CCSDS (Space Packet Protocol, COP-1, CFD), Ground Data System (GDS) integrations

EXPERIENCE

Senior Ground Software Engineer — SpaceX
Mar 2021 – Present

Mission Control Systems | Python, FastAPI, TypeScript, Angular, Kubernetes, InfluxDB

- Designed, scaled, and maintained web-based ground software platforms for Starlink constellation operations, processing over **50 million telemetry packets daily** with sub-second latency.
- Partnered with flight software developers and mission operations teams to translate complex mission requirements into high-performance, Kubernetes-orchestrated microservices.
- Established robust monitoring and alerting frameworks using InfluxDB and Grafana, proactively identifying and mitigating over 99% of data ingestion bottlenecks before affecting operations.
- Spearheaded the integration of CCSDS Space Packet Protocols and COP-1 standards into the telemetry decoding pipeline, ensuring full compliance and reliable command uplink.
- Developed comprehensive E2E testing suites using Playwright, reducing release regression incidents by 35% across mission control UI deployments.

Ground Software Engineer — NASA Jet Propulsion Laboratory (JPL)
Jul 2018 – Feb 2021

Deep Space Network Ground Data Systems | Python, C++, PostgreSQL, Docker, CI/CD

- Engineered high-reliability database architectures and optimized schemas in PostgreSQL for the Deep Space Network, improving query retrieval times by 3x on multi-terabyte data blocks.
- Developed and optimized microservice architectures for spacecraft ground operations, utilizing Python web frameworks and gRPC for high-throughput, low-latency inter-service communication.
- Automated the deployment and configuration of ground data pipelines using Git-based workflows and CI/CD, decreasing deployment times from days to minutes.

PROJECTS

- **OpenSource CCSDS Packet Decoder** (Python, Rust) — Developed a lightning-fast, multi-threaded command and telemetry decoder adhering to CCSDS space data link standards, supporting custom packet schema configurations.
- **Constellation Simulator Dashboard** (Angular, FastAPI, Postgres) — Built a full-stack, real-time satellite constellation telemetry simulator visualizer utilizing WebSockets and InfluxDB time-series storage.

EDUCATION

Massachusetts Institute of Technology (MIT) — Cambridge, MA
Sep 2014 – Jun 2018

B.S. Computer Science & B.S. Aerospace Engineering | GPA 3.9

- Focused on Distributed Systems, Spacecraft Engineering, and Real-Time Software Design.